

(A) energy: only force is conservative central potential, so E conserved.

lin. mom.: has an accelerating force, so $\vec{p}$ not conserved.

ang. mom.: central potential can't apply a torque, so $\vec{L}$ conserved.

(B) $\vec{r}_1 = r \hat{e}_r \rightarrow \dot{\vec{r}} = \dot{r} \hat{e}_r + r \dot{\hat{e}}_r = \dot{r} \hat{e}_r + r \dot{\phi} \hat{e}_\phi$
 $|\dot{\vec{r}}|^2 = \dot{r}^2 + r^2 \dot{\phi}^2$

$$\mathcal{L} = T - V = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\phi}^2) + \lambda r^{-n}$$

$$= \frac{1}{2} m \dot{r}^2 + \lambda r^{-n} + \frac{m r^2 \dot{\phi}^2}{2}$$

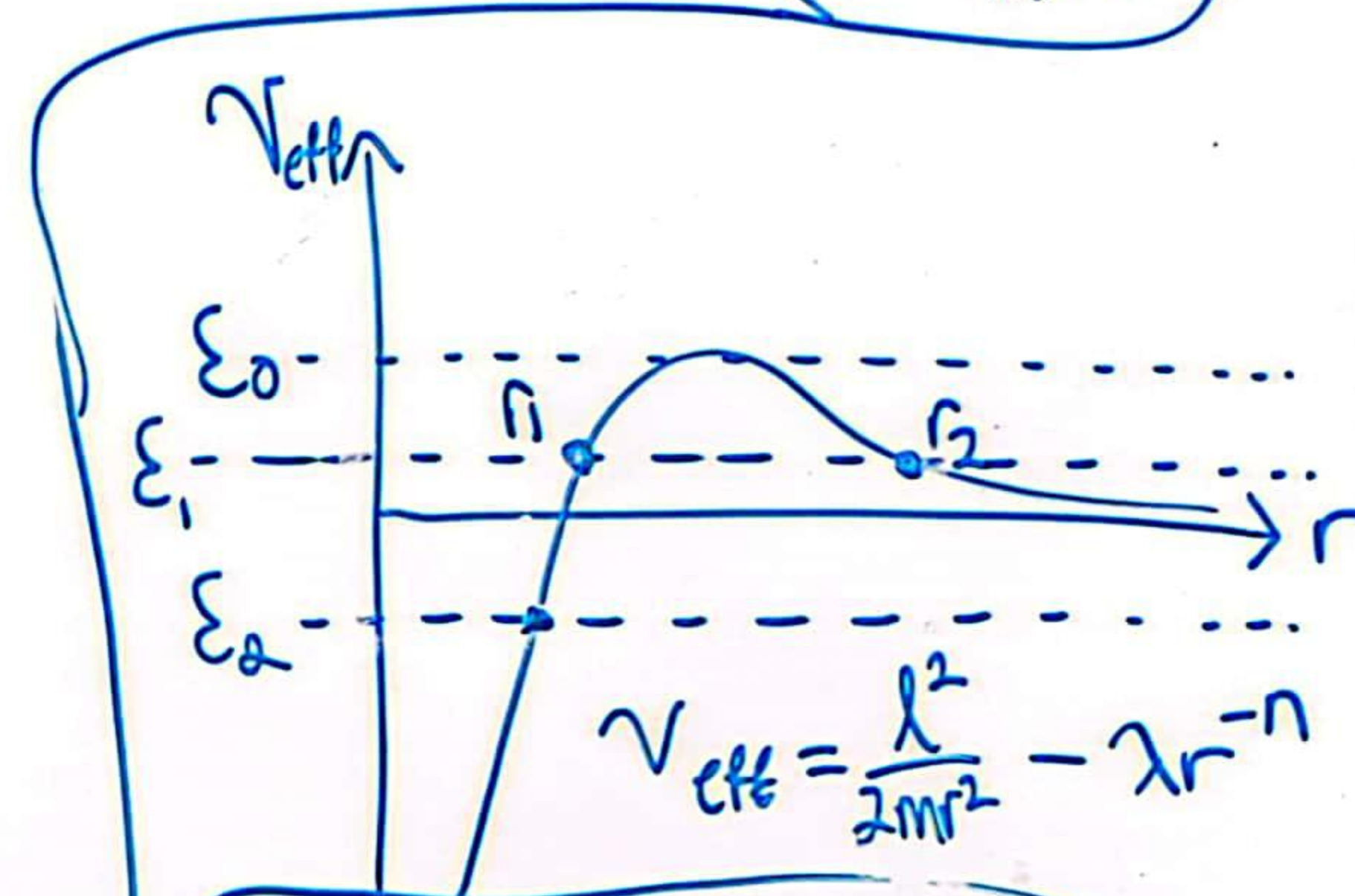
radial: $0 = \partial_t (\partial_{\dot{r}} \mathcal{L}) - \partial_r \mathcal{L} = m \ddot{r} - (-n \lambda r^{-n-1} + m r \dot{\phi}^2)$
 $m \ddot{r} = m r \dot{\phi}^2 - n \lambda r^{-n-1}$

$$m \ddot{r} = m r \frac{l^2}{m^2 r^4} - n \lambda r^{-n-1}$$

azimuthal: $0 = \partial_t (\partial_{\dot{\phi}} \mathcal{L}) - \partial_\phi \mathcal{L}$

$$0 = \partial_t (m r^2 \dot{\phi})$$

$$l = m r^2 \dot{\phi} \rightarrow \dot{\phi} = \frac{l}{m r^2}$$



E_0 - bounded, circular

E_1 - $\begin{cases} r_0 < r_1 & \text{bounded} \\ r_0 > r_2 & \text{unbounded} \end{cases}$

E_2 - bounded

$$\frac{l^2}{m r^3} - n \lambda r^{-n-1}$$

$n > 2$

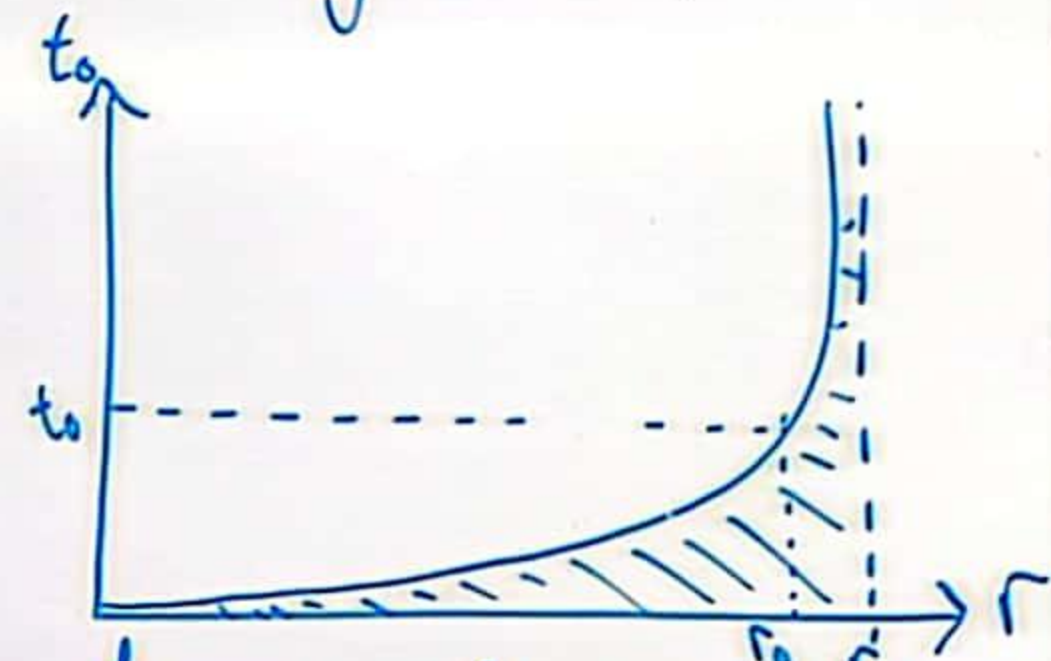
$$\textcircled{c} \quad \mathcal{E} = \mathcal{T} + \mathcal{V} = \frac{m}{2} \dot{r}^2 + \underbrace{\frac{l^2}{2mr^2} - \lambda r^{-n}}_{V_{\text{eff}}}$$

$$\frac{m}{2} \dot{r}^2 = \mathcal{E} - \frac{l^2}{2mr^2} + \lambda r^{-n}$$

$$\frac{dr}{dt} = \pm \left(\frac{2}{m} \left[\mathcal{E} - \frac{l^2}{2mr^2} + \lambda r^{-n} \right] \right)^{1/2}$$

$$t = \pm \int_{r_0}^r \frac{dr}{\left[\frac{2}{m} \left(\mathcal{E} - \frac{l^2}{2mr^2} + \lambda r^{-n} \right) \right]^{1/2}}$$

integrand



expand around $r_1 \rightarrow r = r_1 - \eta, \quad \frac{dr}{d\eta} = -1$

$$V_{\text{eff}}(r) = V_{\text{eff}}(r_1 - \eta) = V_{\text{eff}}(r_1) - \eta V'_{\text{eff}}(r_1) + \dots$$

$$= -\frac{l^2}{2m} (r_1 - \eta)^{-2} + \lambda (r_1 - \eta)^{-n}$$

$$= -\frac{l^2}{2m} r_1^{-2} \left(1 - \left(\frac{\eta}{r_1} \right) \right)^{-2} + \lambda r_1^{-n} \left(1 - \left(\frac{\eta}{r_1} \right) \right)^{-n}$$

$$= -\frac{l^2}{2m r_1^2} \left(1 - 2 \left(-\frac{\eta}{r_1} \right) \right) + \lambda r_1^{-n} \left(1 - n \left(-\frac{\eta}{r_1} \right) \right)$$

$$= \left[-\frac{l^2}{2m r_1^2} + \lambda r_1^{-n} \right] + \left[\frac{l^2}{m r_1^3} \eta + \lambda r_1^{-n-1} \eta \right]$$

for every value $r_0 < r_1$, this is bounded by a rectangle $r_0 \times t_0$, so it is always finite, but r_1 misbehaves.

$$t(\eta) \approx \pm \int_0^\eta \frac{d\eta'}{\left[\frac{2}{m} \left(\mathcal{E} - \frac{l^2}{2m r_1^2} + \lambda r_1^{-n} - \frac{l^2}{m r_1^3} \eta' + \lambda r_1^{-n-1} \eta' \right) \right]^{1/2}}$$

$$= \int \frac{\eta'^{1/2}}{\left[-\frac{l^2}{m r_1^3} + \lambda r_1^{-n-1} \right]^{1/2}} d\eta'$$

$$t(\eta) \approx \frac{1}{2 \left[-\frac{l^2}{2m r_1^3} + \lambda r_1^{-n-1} \right]^{1/2}} \eta^{3/2} \Big|_0^\eta$$

$$= \frac{\eta^{3/2}}{2 \left[\dots \right]^{1/2}} - 0$$

finite time at turning point r_1

$$\varphi = \frac{l}{mr^2} \rightarrow \frac{d\varphi}{dt} = \frac{l}{mr^2} \rightarrow d\varphi = \frac{l}{mr^2} dt$$

$$\varphi = \pm \int \frac{l}{mr^2} \frac{dr}{\left[\frac{2}{m} \left(\mathcal{E} - \frac{l^2}{2mr^2} + \lambda r^{-n} \right) \right]^{1/2}}$$

$$= \pm \int \frac{dr}{r^2 \left[\frac{2}{m} \left(\mathcal{E} - \frac{l^2}{2mr^2} + \lambda r^{-n} \right) \right]^{1/2}}$$

$$dr = d\eta$$

$$r = r_1 - \eta$$

$$\int \frac{dr}{\sqrt{\frac{2}{m} (\mathcal{E} - V_{\text{eff}}(r))}} \approx \int_{r_1-\eta}^{r_1} \frac{dr}{\sqrt{\frac{2}{m} (\mathcal{E} - [V_{\text{eff}}(r_1) - \eta V'_{\text{eff}}(r_1)])}} \approx \int_0^\eta \frac{d\eta}{\sqrt{\frac{2}{m} [V'_{\text{eff}}(r_1) \eta]}}$$

$V_{\text{eff}}(r_1) = \mathcal{E}$

$r_1 - \eta$

$\mathcal{E}$

> 0



OCT 9 2024

$$E = \frac{\mu}{2} \dot{r}^2 + V_{\text{eff}}(r)$$

$$\frac{dr}{dt} = \pm \sqrt{\frac{2}{\mu} (E - V_{\text{eff}}(r))}$$

$$dt = \frac{dr}{\sqrt{\frac{2}{\mu} (E - V_{\text{eff}}(r))}} \Rightarrow t(r)$$

ANGULAR MOMENTUM
CONSERVED

$$l = \mu r^2 \dot{\phi} = \mu r^2 \frac{d\phi}{dt}$$

$\phi(r)$

$$d\phi = \frac{l}{\mu r^2} dt$$

$$d\phi = \pm \frac{l}{\mu r^2} \frac{dr}{\sqrt{\frac{2}{\mu} (E - V_{\text{eff}}(r))}} \Rightarrow \phi(r)$$

① (a) ME: $\vec{F}_g$ is conservative. No drag/dissipative forces. Constraint force does no work.

$P_x, P_y, P_z: X$ $\vec{F}_g$ is external force which has components in $\hat{x}, \hat{y}$, & $\hat{z}$ because of the surface of revolution. (Constraint forces)

$L_x, L_y: X, L_z: \checkmark$

GC: r, ϕ

No net torque bc no net force tangential to the motion.

$$\vec{r}_i = r \cos \phi \hat{x} + r \sin \phi \hat{y} + f(r) \hat{z} = r \hat{r} + f(r) \hat{z}$$

$$\dot{\vec{r}}_i = \frac{\partial \vec{r}_i}{\partial t} + \frac{\partial \vec{r}_i}{\partial r} \dot{r} + \frac{\partial \vec{r}_i}{\partial \phi} \dot{\phi}$$

$$z = f(r)$$

$$\dot{z} = f'(r) \dot{r}$$

ALTERNATIVE

$$\vec{r} = r \hat{r} + f(r) \hat{z}$$

$$\dot{\vec{r}} = \dot{r} \hat{r} + r \dot{\phi} \hat{\phi} + f'(r) \dot{r} \hat{z}$$

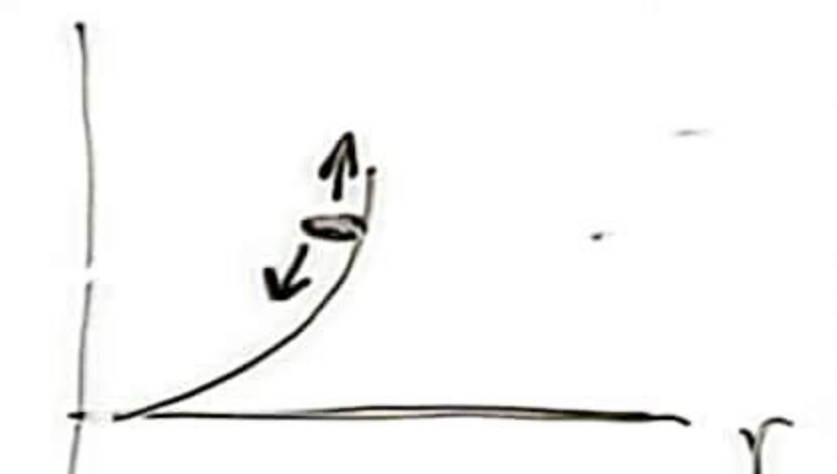
$$= \dot{r} \hat{r} + r \dot{\phi} \hat{\phi} + f'(r) \dot{r} \hat{z}$$

$$|\dot{\vec{r}}|^2 = \dot{r}^2 + r^2 \dot{\phi}^2 + f'(r)^2 \dot{r}^2$$

Constraints z

$$|c|=1 \Rightarrow z = f(r)$$

$$n=2$$



$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = \frac{\partial L}{\partial \phi}$$

$$\frac{\partial L}{\partial \phi} = 0 \Rightarrow \phi \text{ is cyclic} \Rightarrow P_\phi \text{ is conserved}$$

$$\frac{\partial L}{\partial \dot{\phi}} = m r^2 \dot{\phi} = P_\phi = l = \text{constant}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = \frac{\partial L}{\partial r}$$

$$\frac{\partial L}{\partial \dot{r}} = m \dot{r} + m f'(r)^2 \dot{r} = \dot{r} m (1 + f'(r)^2)$$

$$|\dot{\vec{r}}|^2 = \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i = \dot{r}^2 + r^2 \dot{\phi}^2 + f'(r)^2 \dot{r}^2$$

$$L = T - V = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2 + f'(r)^2 \dot{r}^2) - m g f(r)$$

$$L = \frac{m}{2} (r^2 \dot{\phi}^2 + \dot{r}^2 (1 + f'(r)^2)) - m g f(r)$$

$$\ddot{r} = \frac{r\dot{\psi}^2 + \dot{r}^2 f'(r)f''(r) - g f'(r) - 2\dot{r}^2 f'(r)f''(r)}{(1+f'(r)^2)} = \frac{r\dot{\psi}^2 - g f'(r) - \dot{r}^2 f'(r)f''(r)}{(1+f'(r)^2)}$$

For steady circular orbit, r must be fixed.

$$r\dot{\psi}^2 = \frac{l^2}{m^2 r^3}$$

$$\dot{r} = \ddot{r} = 0 \Rightarrow r = r_0$$

$$\ddot{\eta} = -c\eta$$

Remember $Mr_0^2\dot{\psi} = l$ is constant

$$0 = r_0 \left(\frac{l^2}{m^2 r_0^4} \right) - g f'(r_0)$$

$$\boxed{f'(r_0) = \frac{l^2}{gm^2 r_0^3}}$$

$$r = r_0 + \eta$$

$$\dot{r} = \dot{\eta}$$

Taylor Expand around $r = r_0 + \eta$

$$f'(r) = f'(r_0 + \eta) = f'(r_0) + \eta f''(r_0)$$

Make substitution into eq. of motion

$$\dot{r}'(1+f'(r)^2) = \frac{l^2}{m^2 r^3} - g f'(r) - \dot{r}^2 f'(r)f''(r)$$

$$\dot{\eta}'(1+f'(r_0+\eta)^2) = \frac{l^2}{m^2 (r_0+\eta)^3} - g f'(r_0+\eta)$$

$$\frac{1}{(r_0+\eta)^3} = \frac{1}{r_0^3} \left(1 + \frac{\eta}{r_0}\right)^{-3} = \frac{1}{r_0^3} \left(1 - \frac{3\eta}{r_0}\right)$$

$$\dot{\eta}'(1+f'(r_0)^2) = -g(3f'(r_0) + r_0 f''(r_0))\eta$$

$$\ddot{\eta} = - \left(\frac{g(3f'(r_0) + r_0 f''(r_0))}{1+f'(r_0)^2} \right) \eta$$

$$(1+x)^2 \approx 1+2x$$

Binomial
Approximation

$$\omega = \sqrt{c}$$